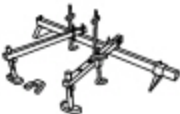


Engine Support Insulators

 [Ford Ecat Electronic Parts Catalog](#)

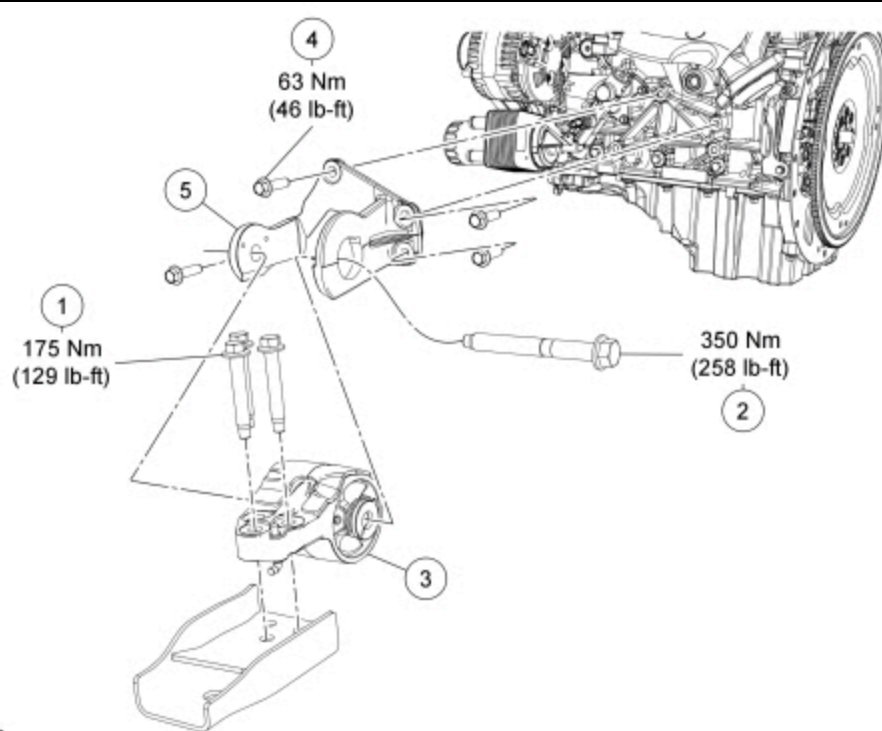
Special Tool(s)

 ST3101-A	Bar, Engine Spreader 303-1454
 ST2976A	Eye, Engine Lift 303-1245
 ST2176-A	Support Bar, Engine 303-F070

Material

Item	Specification
Motorcraft® Threadlock 262 TA-26	WSK-M2G351-A6

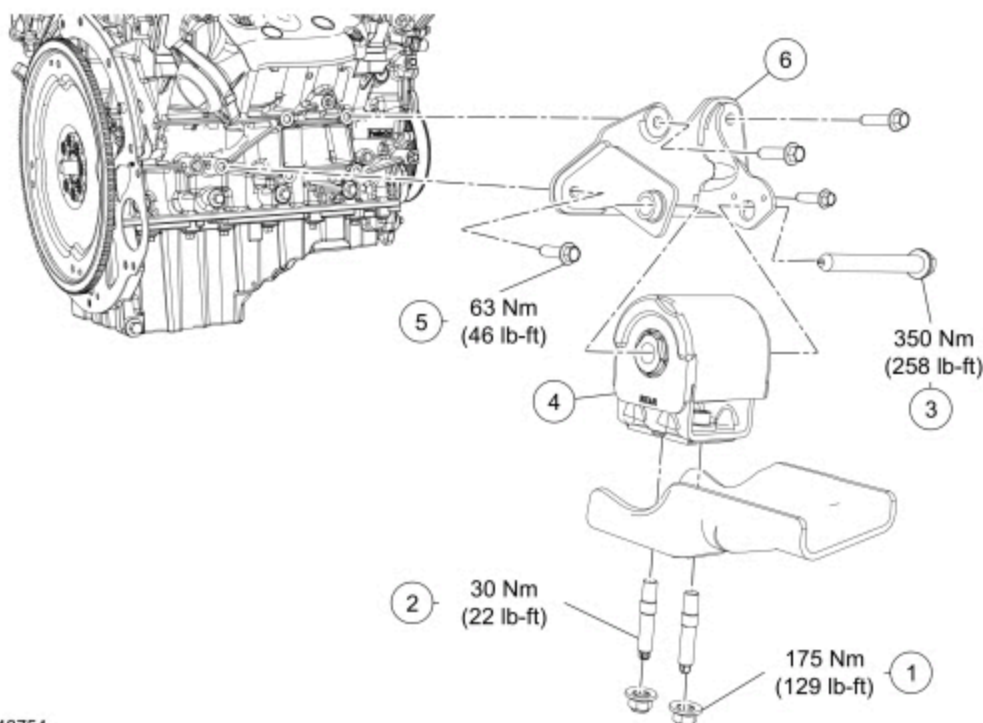
Engine Support Insulator — LH



N0121739

Item	Part Number	Description
1	W713244 W713244	LH engine support insulator-to-frame bolt (3 required)
2	W715211 W715211	LH engine support insulator through bolt
3	6038 6038	LH engine support insulator
4	W714735 W714735	LH engine support insulator-to-cylinder block bracket bolt (4 required)
5	6038 6038	LH engine support insulator-to-cylinder block bracket

Engine Support Insulator — RH



N0148754

Item	Part Number	Description
1	W707251 W707251	RH engine support insulator nut (2 required)
2	—	RH engine support insulator stud bolt (part of 6038) (2 required)
3	—	RH engine support insulator through bolt (part of 6038)
4	6038 6038	RH engine support insulator
5	W714735 W714735	RH engine support insulator-to-cylinder block bracket bolt (4 required)
6	—	RH engine support insulator-to-cylinder block bracket (part of 6038)

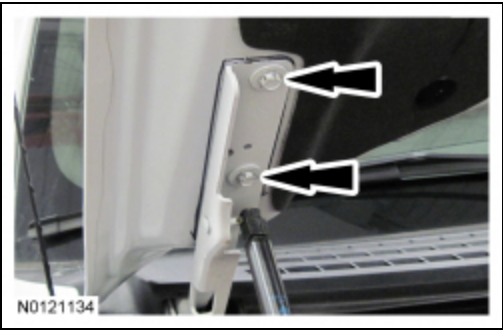
Removal

All engine support insulator

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. **NOTE:** *Index-mark the hood hinge location to aid in hood installation.*

NOTE: *RH shown, LH similar.*

Remove the 4 bolts (2 shown) and the hood.



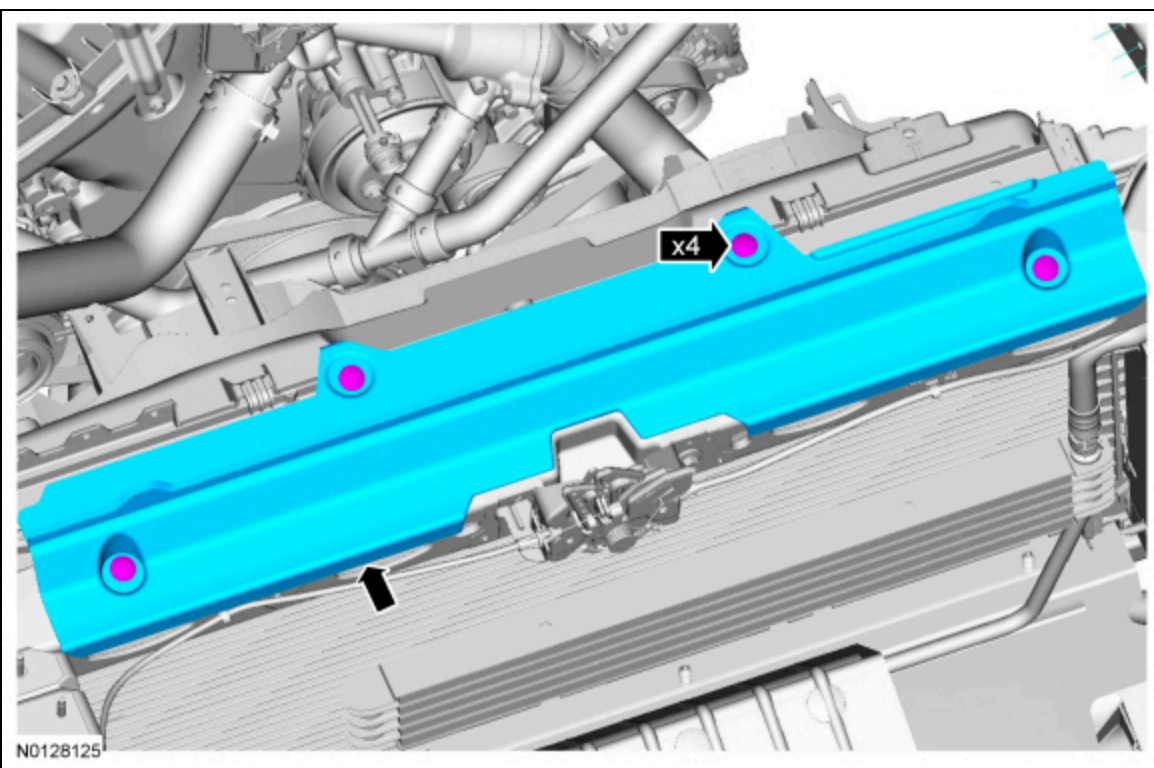
3. Remove the upper intake manifold. For additional information, refer to Upper Intake Manifold in this section.
4. Remove the cowl panel grille. For additional information, refer to Section 501-02.
5. Remove the 6 screws and release the 2 cowl panel clips and remove the cowl panel.



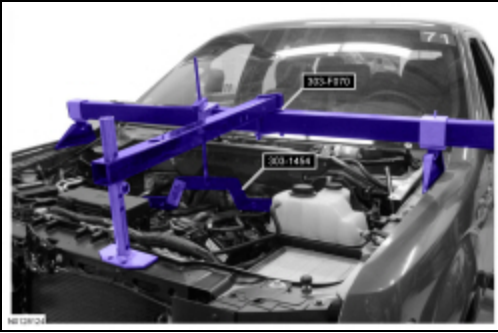
6. Install the LH engine lift eye.



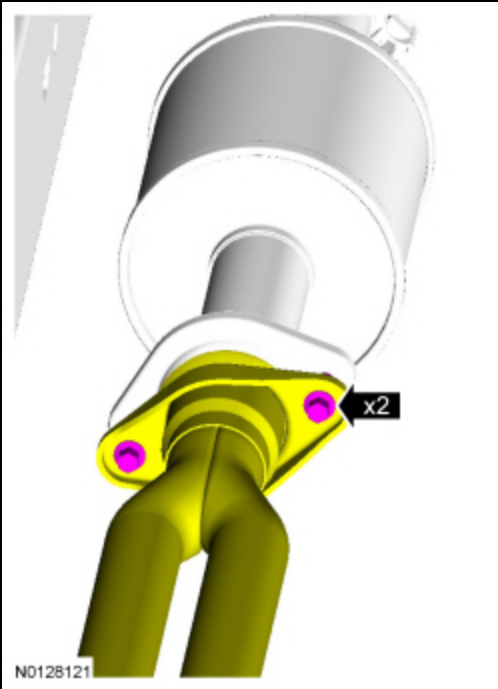
7. Remove the 4 pushpins and the upper radiator air deflector.



8. Install the Engine Support Bar.

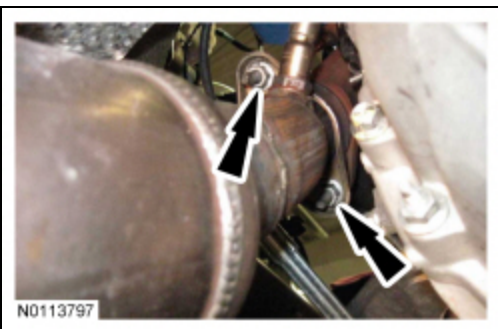


9. If equipped, remove the front driveshaft. For additional information, refer to Section 205-01.
10. Remove the 2 exhaust Y-pipe dual catalytic converter-to-exhaust intermediate pipe bolts and separate.
 - Discard the 2 bolts.



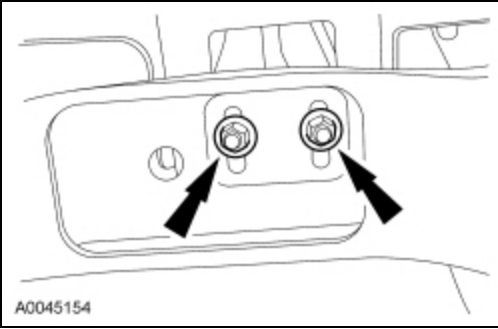
11. **NOTE:** *LH shown, RH similar.*

Remove and discard the 2 LH and RH catalytic converter-to-exhaust manifold nuts.



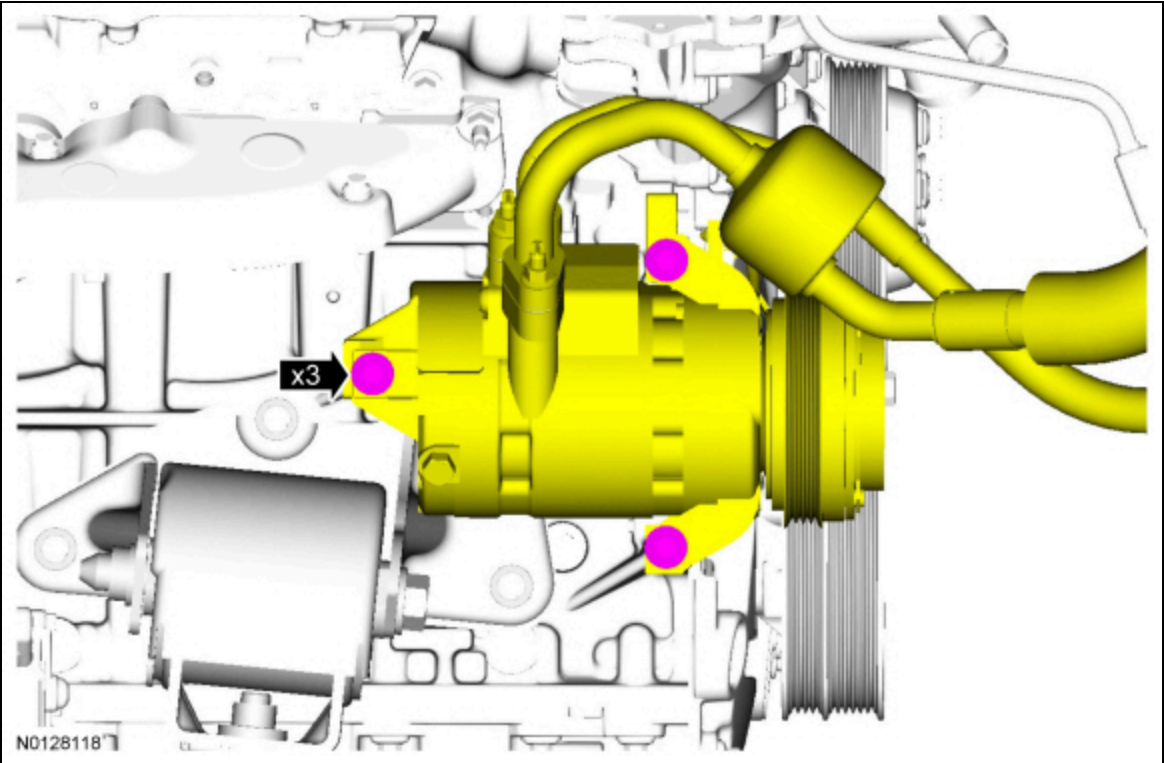
12. **NOTICE:** Only use hand tools when removing the transmission mount-to-crossmember nuts or damage to the transmission mount can occur.

Loosen the 2 transmission mount-to-crossmember nuts.



RH engine support insulator

13. Remove the Air Conditioning (A/C) compressor belt. For additional information, refer to Section 303-05.
14. Remove the 3 bolts and position the A/C compressor aside.

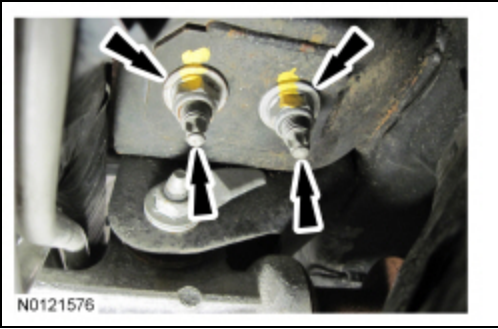


15. Remove the starter. For additional information, refer to Section 303-06.
16. **NOTICE: Only use hand tools when loosening or tightening the engine support insulator through bolts or damage to the engine support insulator-to-cylinder block bracket can occur.**
Remove and discard the RH and LH engine support insulator through bolts.
17. **NOTICE: Only use hand tools when removing or installing the engine support insulator nuts or damage to the engine support insulator can occur.**

NOTICE: Only use hand tools when removing or installing the engine support insulator studs or damage to the engine support insulator can occur.

NOTE: *The 2 engine support insulator studs may come off with the nuts.*

Remove and discard the 2 RH engine support insulator nuts. Remove the engine support insulator studs.



18. Raise the engine approximately 30 mm (1.18 in).

19. Remove the RH engine support insulator.

20. Remove the 4 bolts and the RH engine support insulator-to-cylinder block bracket.

LH engine support insulator

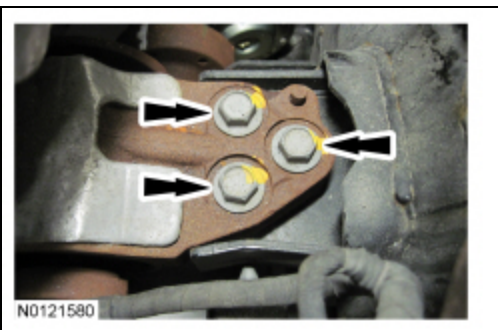
21. **NOTICE: Only use hand tools when loosening or tightening the engine support insulator through bolts or damage to the engine support insulator-to-cylinder block bracket can occur.**

Remove and discard the LH and RH engine support insulator through bolts.

22. Raise the engine approximately 30 mm (1.18 in).

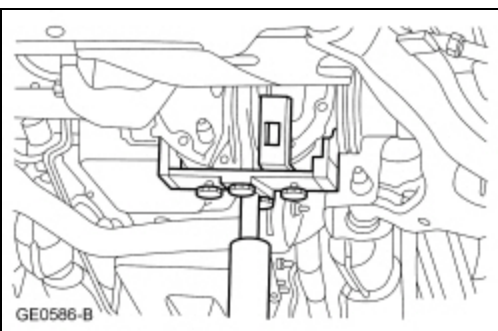
23. **NOTICE: Only use hand tools when removing or installing the engine support insulator-to-frame bolts or damage to the engine support insulator-to-frame nut plate can occur.**

Remove the 3 LH engine support insulator-to-frame bolts.

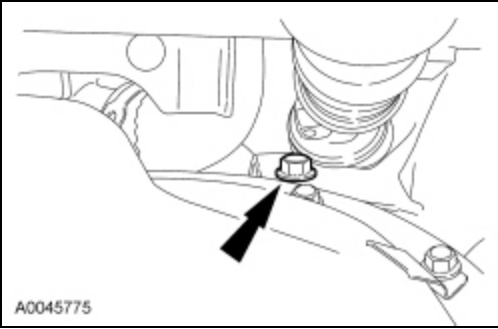


Four-Wheel Drive (4WD) — LH engine support insulator

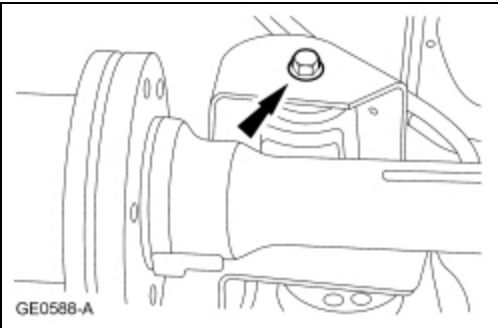
24. Position a suitable hydraulic jack under the front axle. Securely strap the jack to the axle.



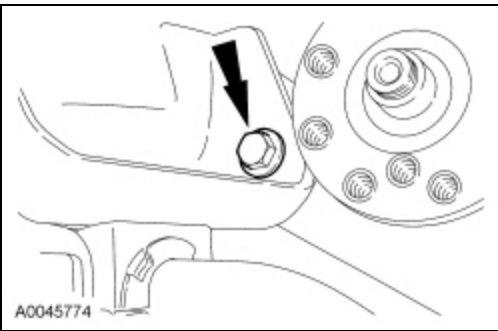
25. Remove the upper front axle carrier mounting bushing bolt.



26. Remove the axle shaft housing carrier bushing bolt.



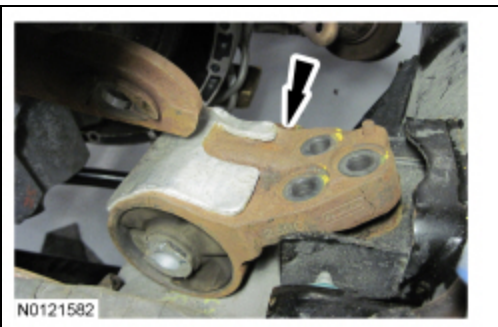
27. Remove the lower front axle carrier mounting bushing bolt.



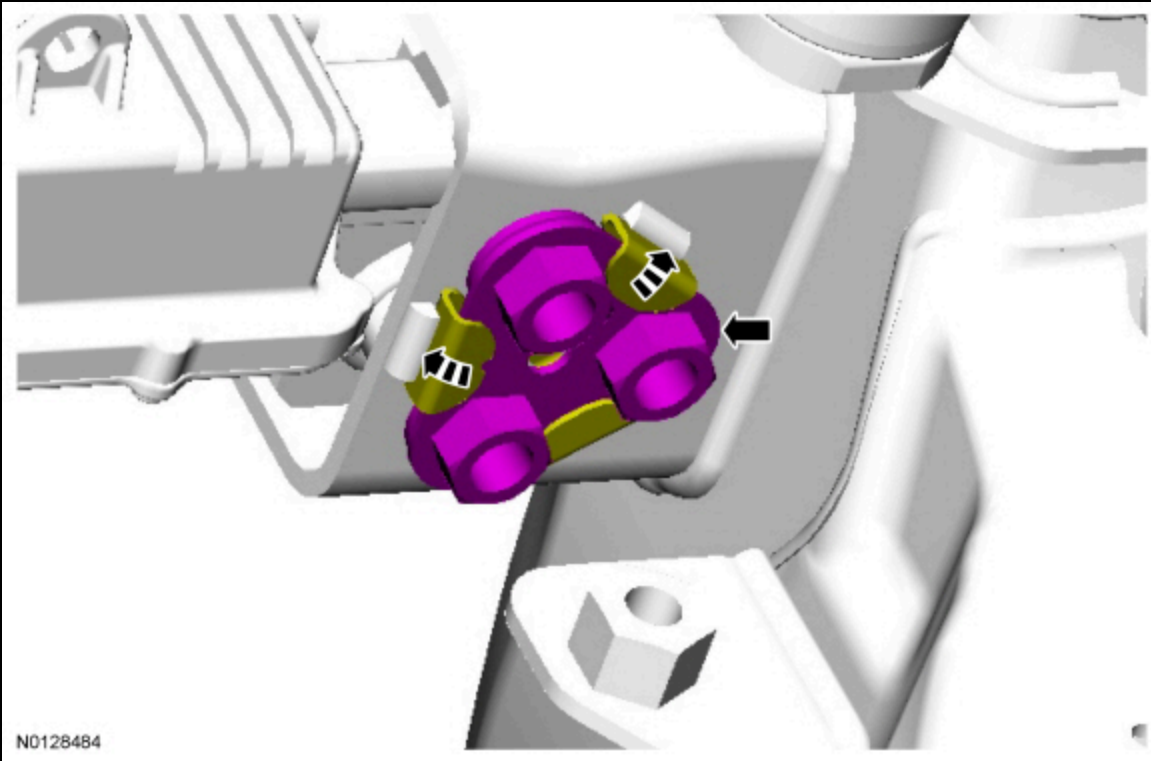
28. Lower the front axle to allow clearance to the LH engine support insulator.

LH engine support insulator

29. Remove the LH engine support insulator.



30. Inspect the engine support insulator-to-frame nut plate for thread damage. If the nut plate is damaged, bend 2 tabs back and remove the nut plate. Replace the nut plate with service part number 56190 and bend the 2 tabs back to the original position.

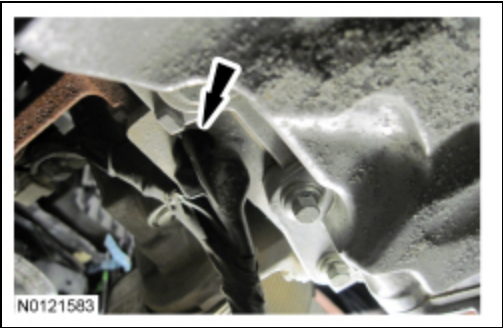


NOTE: If necessary, the following 3 steps are to remove the LH engine support insulator-to-cylinder block bracket.

31. Detach the generator wiring harness from the engine front cover.
 - Detach the generator wiring harness retainer from the oil filter adapter.



32. Detach the generator wiring harness retainer from the lower LH side of the engine block.



33. Remove the 4 bolts and the LH engine support insulator-to-cylinder block bracket.

Installation

LH engine support insulator

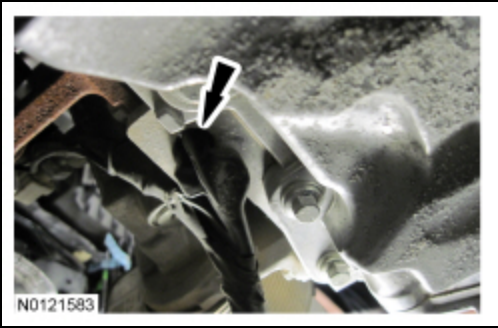
1. Clean the engine support insulator-to-cylinder block and engine support insulator-to-frame mating surfaces of any dirt or foreign material prior to installation.

NOTE: If removed, the following 3 steps are to install the LH engine support insulator-to-cylinder block bracket.

2. Install the LH engine support insulator-to-cylinder block bracket and install the 4 bolts.

- Tighten to 63 Nm (46 lb-ft).

3. Attach the generator wiring harness retainer to the lower LH side of the engine block.



4. Attach the generator wiring harness to the engine front cover.

- Attach the generator wiring harness retainer to the oil filter adapter.



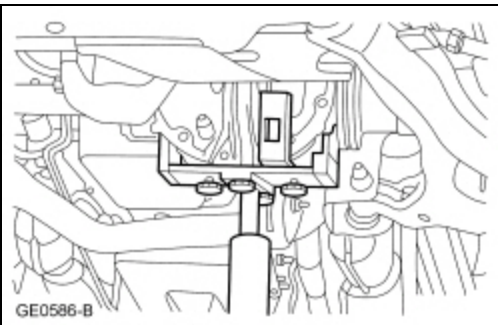
5. Position the LH engine support insulator and hand start the 3 new engine support insulator-to-frame bolts.

- Do not torque at this time.



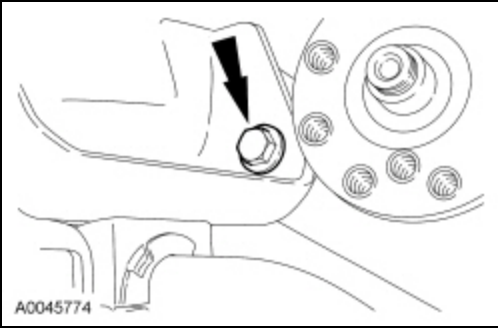
4WD — LH engine support insulator

6. Raise the front axle carrier into position.

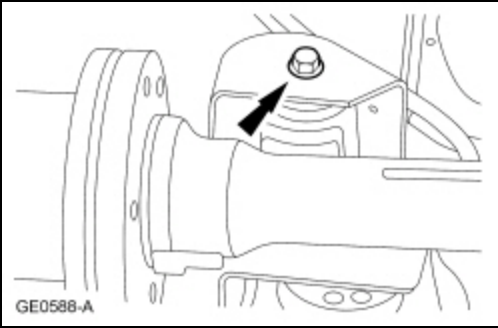


7. Install the lower front axle carrier mounting bushing bolt.

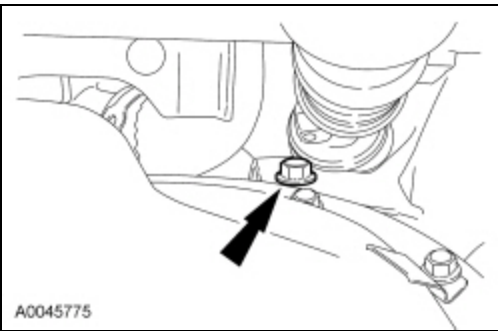
- Tighten to 115 Nm (85 lb-ft).



8. Install the axle shaft housing carrier bushing bolt.
 - Tighten to 115 Nm (85 lb-ft).

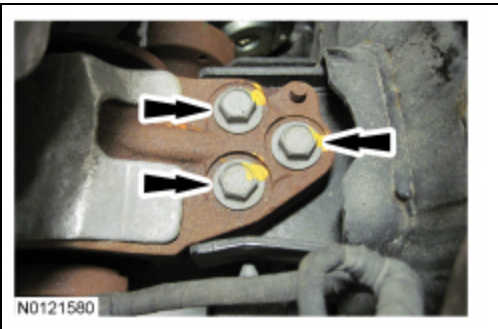


9. Install the upper front axle carrier mounting bushing bolt.
 - Tighten to 115 Nm (85 lb-ft).



LH engine support insulator

10. Tighten the LH engine support insulator-to-frame bolts to 175 Nm (129 lb-ft).



11. Lower the engine to the installed position.
12. **NOTICE: Only use hand tools when loosening or tightening the engine support insulator through bolts or damage to the engine support insulator-to-cylinder block bracket can occur.**

- Install the new RH and LH engine support insulator through bolts.
- Tighten to 350 Nm (258 lb-ft).

RH engine support insulator

13. Clean the engine support insulator-to-cylinder block and engine support insulator-to-frame mating surfaces of any dirt or foreign material prior to installation.

14. Position the RH engine support insulator-to-cylinder block bracket and install the 4 bolts.

- Tighten to 63 Nm (46 lb-ft).

15. **NOTICE: Only use hand tools when installing the engine support insulator studs or damage to the engine support insulator can occur.**

NOTE: *Apply threadlock to the stud threads prior to installation.*

Position the RH engine support insulator into the vehicle and install the 2 studs.

- Tighten to 30 Nm (22 lb-ft).

16. **NOTICE: Only use hand tools when installing the RH engine support insulator nuts or damage to the engine support insulator can occur.**

Install the 2 new RH engine support insulator nuts.

- Tighten to 175 Nm (129 lb-ft).

17. Lower the engine to the installed position.

18. **NOTICE: Only use hand tools when loosening or tightening the engine support insulator through bolts or damage to the engine support insulator-to-cylinder block bracket can occur.**

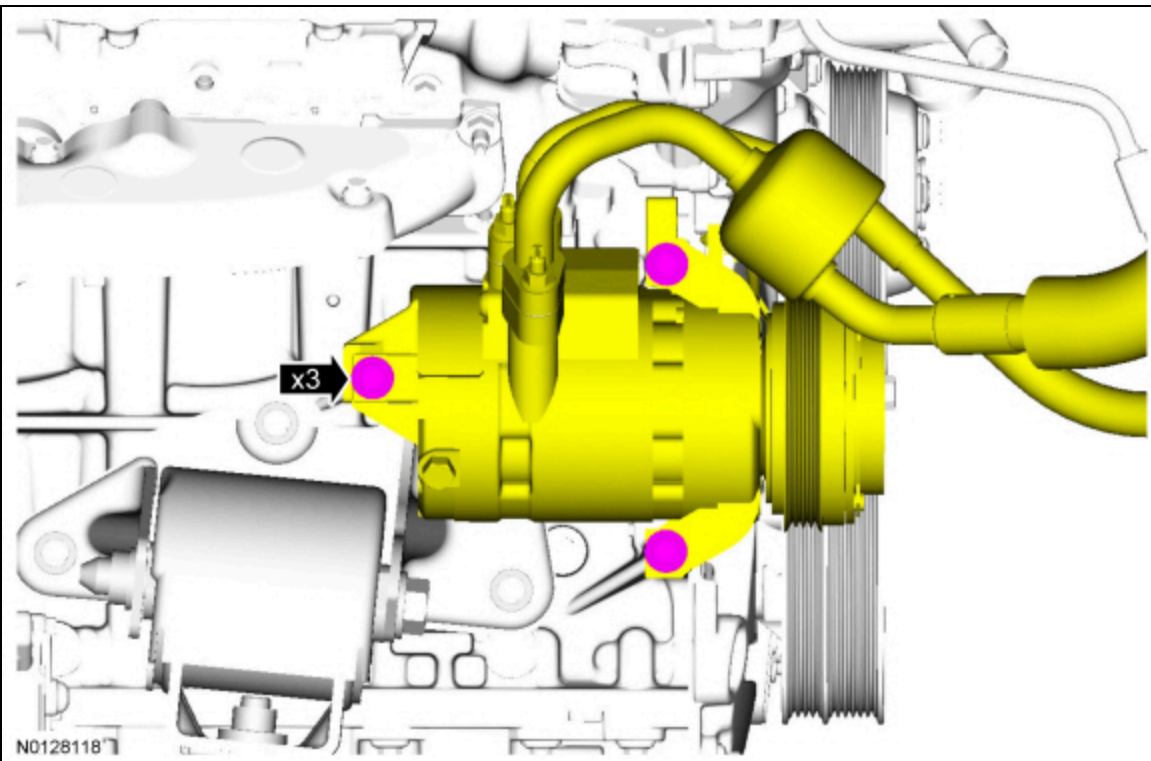
Install the new RH and LH engine support insulator through bolts.

- Tighten to 350 Nm (258 lb-ft).

19. Install the starter. For additional information, refer to Section 303-06.

20. Position the A/C compressor and install the 3 bolts.

- Tighten to 25 Nm (18 lb-ft).

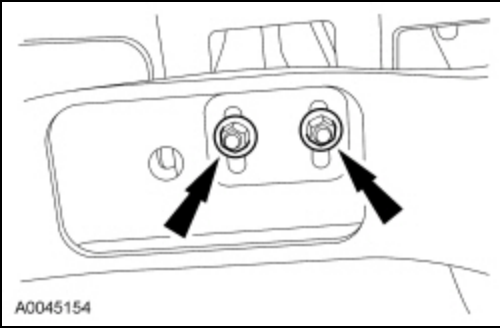


21. Install the A/C compressor belt. For additional information, refer to Section 303-05.

All vehicles

22. **NOTICE:** Only use hand tools when installing the transmission mount-to-crossmember nuts or damage to the transmission mount can occur.

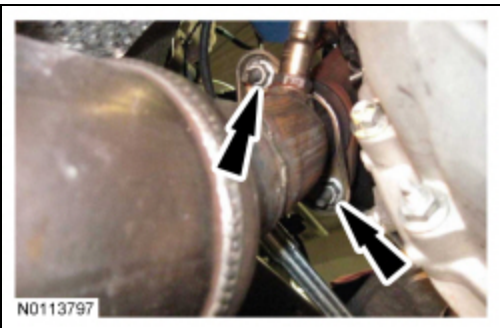
Tighten the transmission mount-to-crossmember nuts to 103 Nm (76 lb-ft).



23. **NOTE:** LH shown, RH similar.

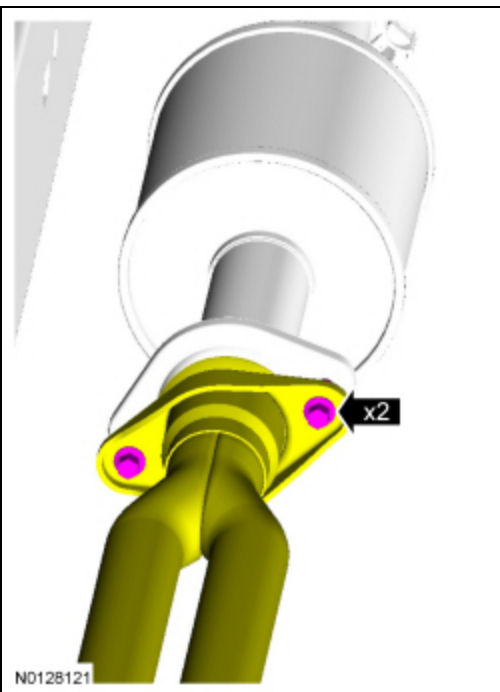
Install the new 2 LH and RH catalytic converter-to-exhaust manifold nuts.

- Tighten to 45 Nm (35 lb-ft).



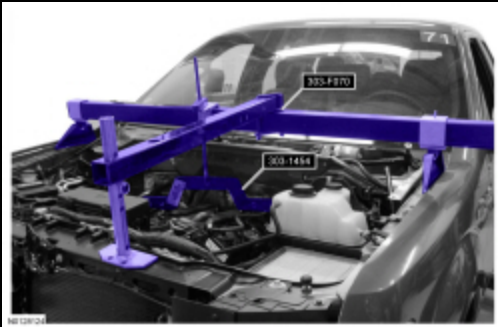
24. Position the exhaust Y-pipe dual catalytic converter-to-exhaust intermediate pipe and install the 2 new bolts.

- Tighten the inner exhaust Y-pipe dual catalytic converter-to-exhaust intermediate pipe bolt to 63 Nm (46 lb-ft).
- Tighten the outer exhaust Y-pipe dual catalytic converter-to-exhaust intermediate pipe bolt to 63 Nm (46 lb-ft).

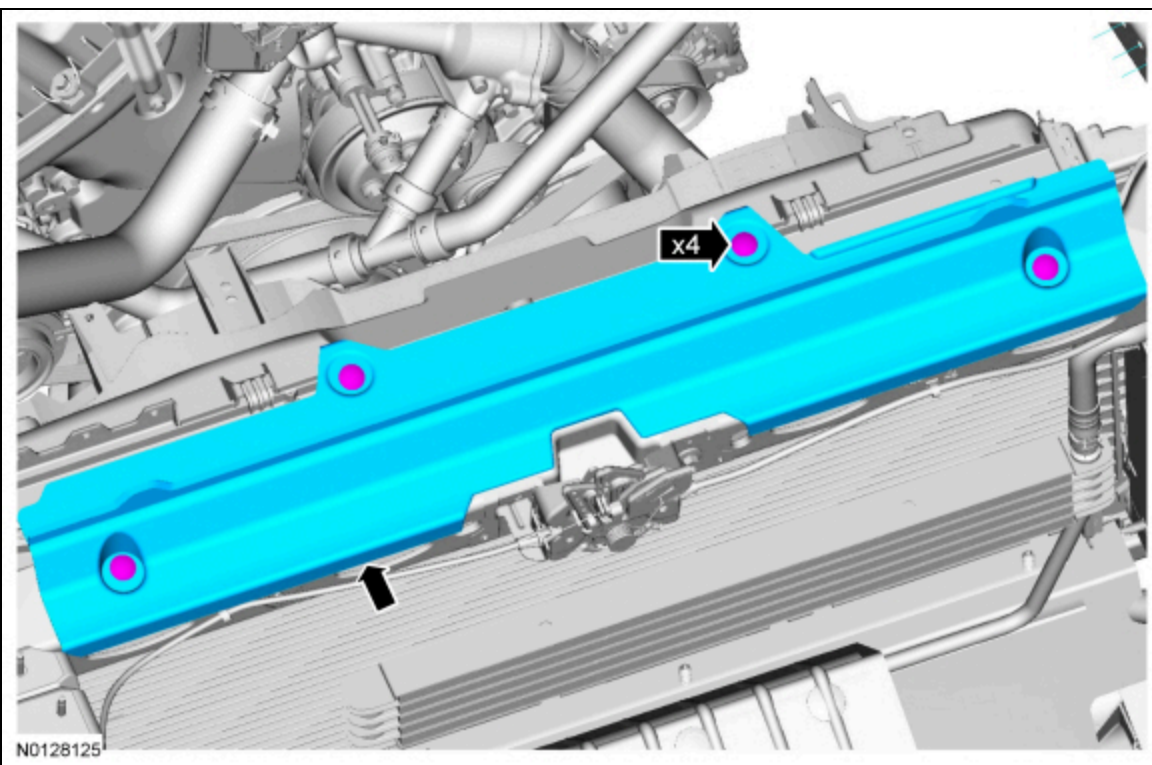


25. If equipped, install the front driveshaft. For additional information, refer to Section 205-01.

26. Remove the Engine Support Bar.



27. Install the upper radiator air deflector and the 4 pushpins.



28. Remove the LH engine lift eye.



29. Install the cowl panel and attach the 2 clips and install the 6 screws.



30. Install the cowl panel grille. For additional information, refer to Section 501-02.

31. Install the upper intake manifold. For additional information, refer to Upper Intake Manifold in this section.

32. **NOTE:** *Index-mark the hood hinge location to aid in hood installation.*

NOTE: *RH shown, LH similar.*

Position the hood and install the 4 bolts.

- Tighten to 12 Nm (106 lb-in).

